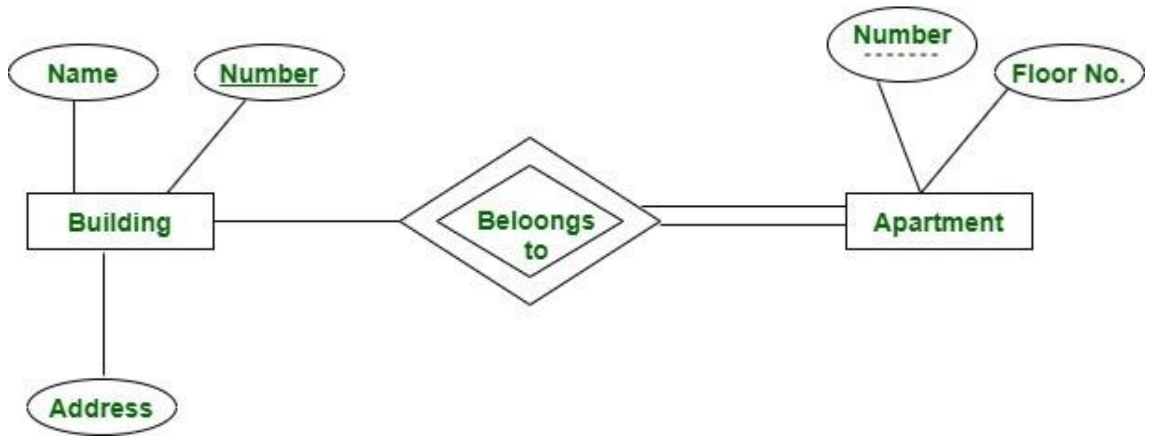
# Chapter 3 continued

## Partial Key :

- The set of attributes that are used to uniquely identify a weak entity set is called the **Partial key**. Only a bunch of the tuples can be identified using the partial keys. The partial Key of the weak entity set is also known as a **discriminator.**

- It is just a part of the key as only a subset of the attributes can be identified using it. It is partially unique and can be combined with other strong entity set to uniquely identify the tuples.

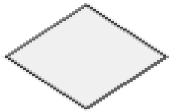
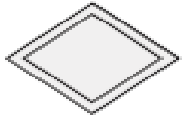
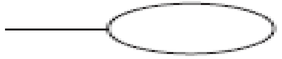
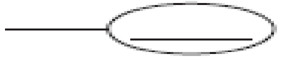
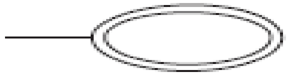
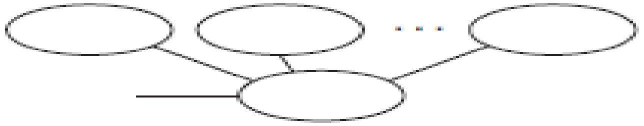


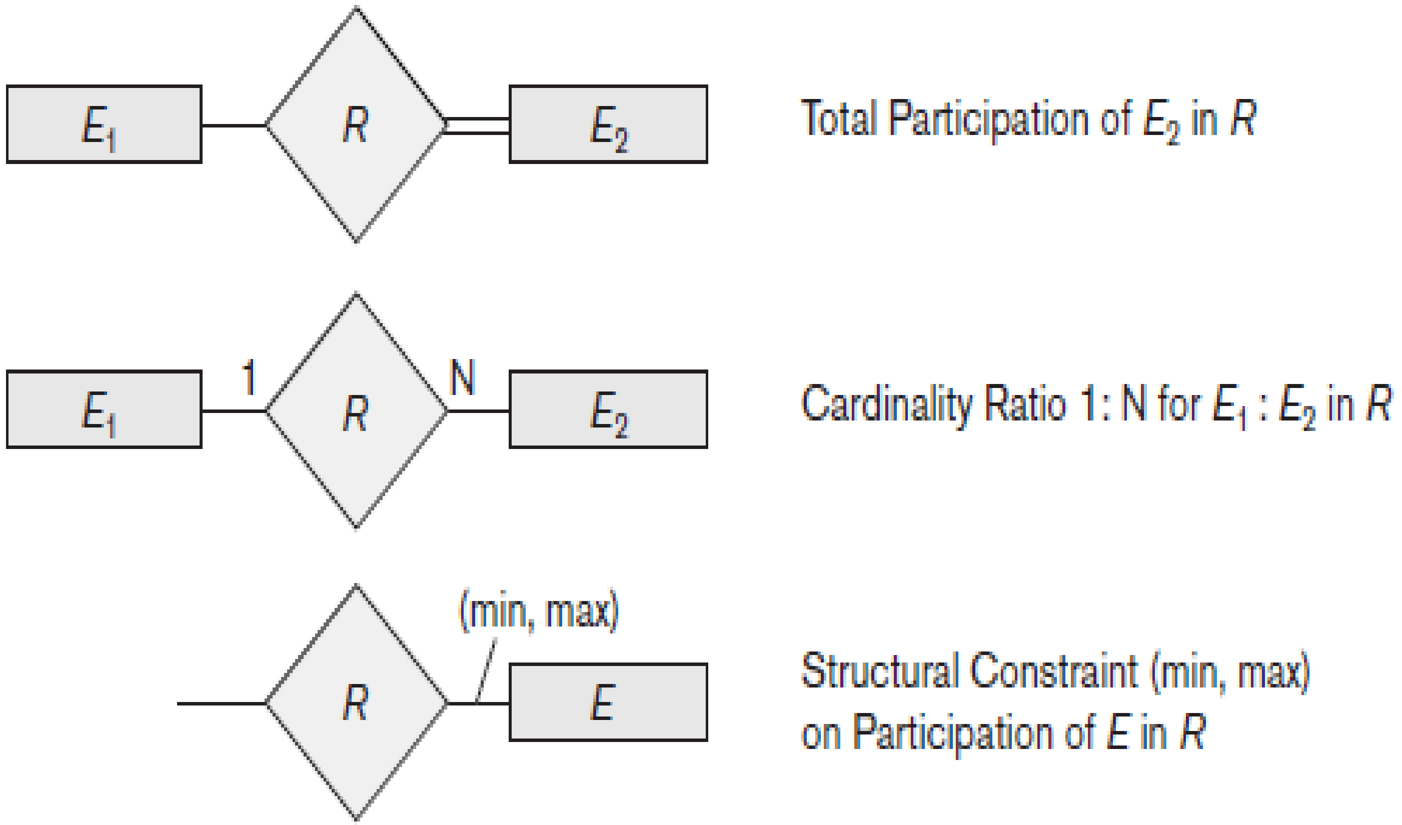

Here we have an apartment as a weak entity and building as a strong entity type connected via ‘belongs to’ relationship set. **Apartment number** is not globally unique i.e. more than one apartment may have same number globally but it is unique for a particular building since a building may not have same apartment number. Thus apartment number cannot be primary key of entity Apartment but it is a partial key shown with a dashed line.

## Strong entity vs Weak entity

| Strong entity | Weak entity |
|---|---|
| Strong entity always has a primary key. | It will not have a primary key but it has partial discriminator key. |
| It is not dependent on any other entity. | Weak entity is dependent on the strong entity. |
| Represented by a single rectangle. | Represented by double rectangle. |
| Relationship between two strong entities is represented by a single diamond. | Relationship between a strong entity and the weak entity is represented by double Diamond. |
| A strong entity may or may not have total participation. | It has always total participation. |

# Summary of the notation for ER diagrams

| Symbol | Meaning |
|---|---|
|  | Entity |
|  | Weak Entity |
|  | Relationship |
|  | Indentifying Relationship |
|  | Attribute |
|  | Key Attribute |
|  | Multivalued Attribute |
|  | Composite Attribute |
|  | Derived Attribute |
|   | Weak key attribute |


E1
R
E2
Total Participation of E2 in R
E1
1
R
N
E2
Cardinality Ratio 1: N for E1 : E2 in R
(min, max)
R
E
Structural Constraint (min, max)
on Participation of E in R

## Alternative (min, max) notation for relationship structural constraints:

- Associates a pair of integer numbers (min, max) with each participation of an entity type in a relationship type where

$$0 \leq min \leq max \text{ and } max \geq 1.$$

- Specifies that each entity e in E participates in at least *min* and at most *max* relationship instances in R at any point in time.
- Min=0 implies partial participation, whereas min>0 implies total participation.
- min=0, max=n (signifying no limit)
- Must have min≤max, min≥0, max ≥1

- **Eg:**
  - A department has exactly one manager and an employee can manage at most one department.
    - Specify (0,1) for participation of EMPLOYEE in MANAGES
    - Specify (1,1) for participation of DEPARTMENT in MANAGES

- An employee can work for exactly one department but a department can have any number of employees.
  - Specify (1,1) for participation of EMPLOYEE in WORKS_FOR
  - Specify (0,n) for participation of DEPARTMENT in WORKS_FOR

Fig : a)

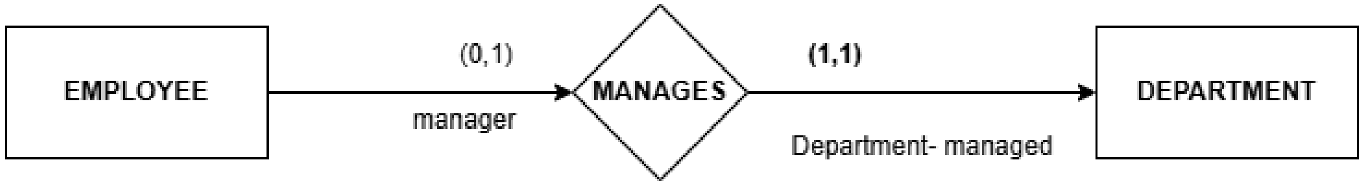
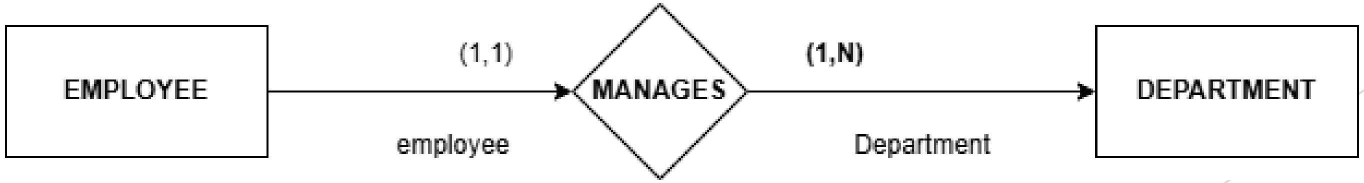



Fig : b)



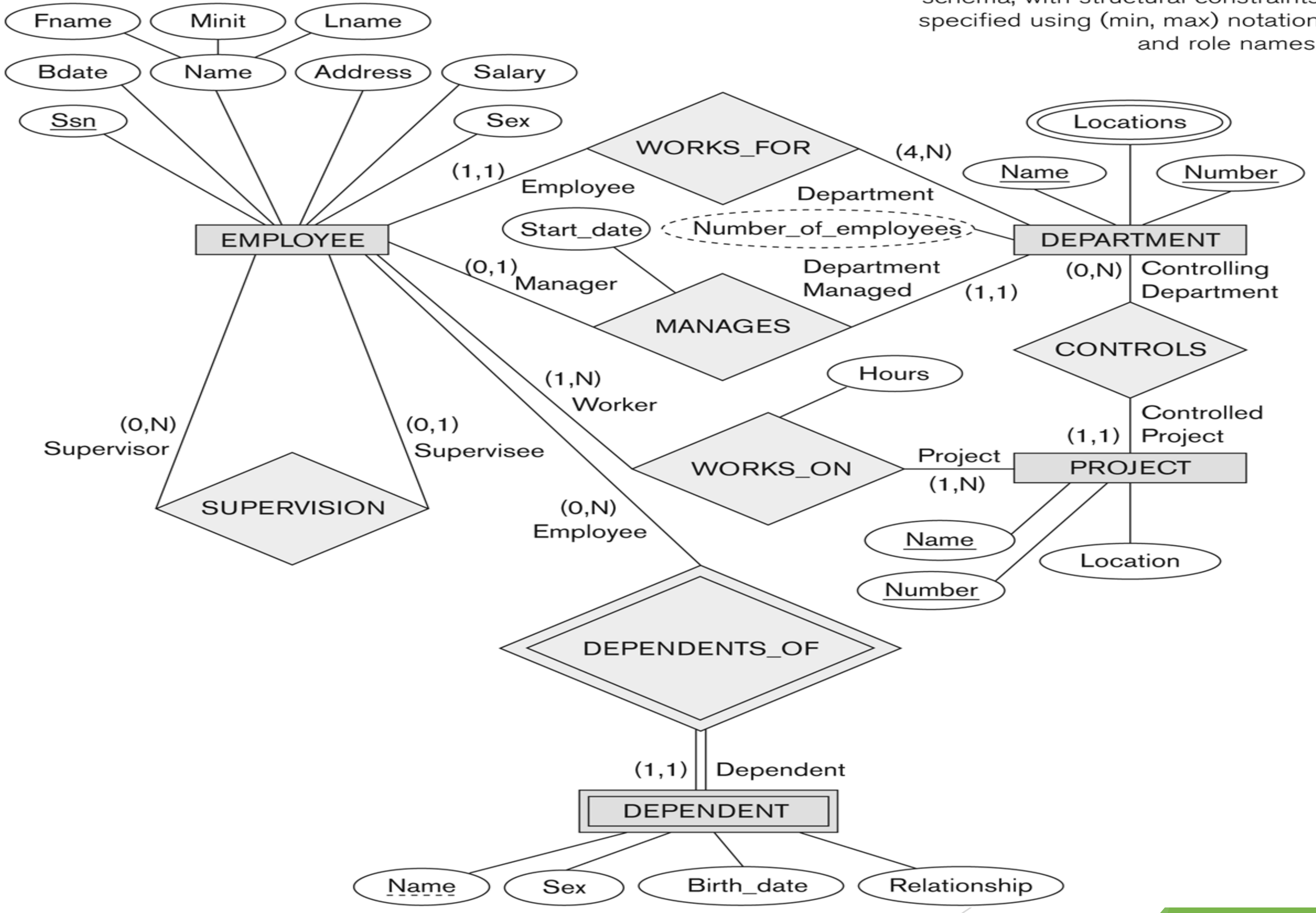



**Figure 3.15**
ER diagrams for the company schema, with structural constraints specified using (min, max) notation and role names.

# ER Diagram Naming Conventions and Design Choices

## Naming Convention in ER diagram:

1) We should choose names that convey, as much as possible, the meanings attached to the different constructs in the schema.

2) We choose to use singular names for entity types, rather than plural ones, because the entity type name applies to each individual entity belonging to that entity type.

3) In our ER diagrams, we will use the convention that entity type and relationship type names are in uppercase letters, attribute names have their initial letter capitalized, and role names are in lowercase letters.

4) Choosing binary relationship names to make the ER diagram of the schema readable from left to right and from top to bottom.

**Example:**

1) Imagine you have a database for a library. Instead of naming an entity type just "Books," you might choose "Book" to convey that each record in that entity type represents an individual book.

2) Continuing with the library example, if you have an entity type for authors, use "Author" instead of "Authors" because each record represents a single author.

3) In your ER diagram, if you have an entity type for "Book," write it as "BOOK." If there's an attribute like "Title," capitalize the initial letter, and if there's a role like "written by," use lowercase letters.

4) Consider a binary relationship in the library schema between "Book" and "Author." Naming it "Authored by" helps the ER diagram flow logically from left to right and from top to bottom, making it easier to understand.

## Design Choices for ER

**A schema design process is an iterative refinement process. Some of the refinements include the following:**

- A concept may be first modeled as an attribute and then refined into relationship because it is determined that the attribute is a reference to another entity type.

DEPARTMENT
Name, Number, {Locations}, Manager, ManagerStartDate

PROJECT
Name, Number, Location, ControllingDepartment

EMPLOYEE
Name (FName, MInit, LName), SSN, Sex, Address, Salary, BirthDate, Department, Supervisor, {WorksOn (Project, Hours)}

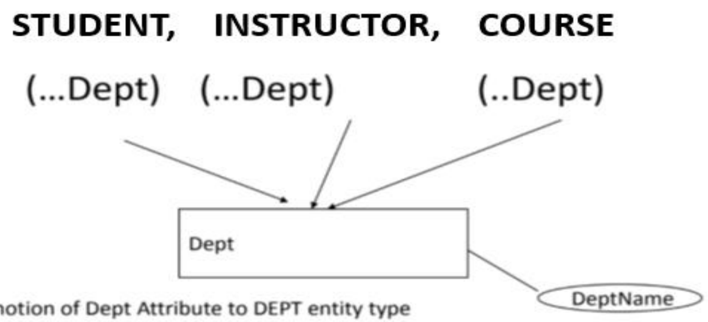
DEPENDENT
Employee, DependentName, Sex, BirthDate, Relationship

**Initial conceptual design**

**Eg :** A "Department" has a "Manager," initially modeled as an attribute. Later, it might be represented as a relationship because a Manager could be a separate "Employee" entity.

The schema presented shows examples of entities like Department, Project, Employee, and Dependent with their attributes, illustrating how initial design choices can evolve.

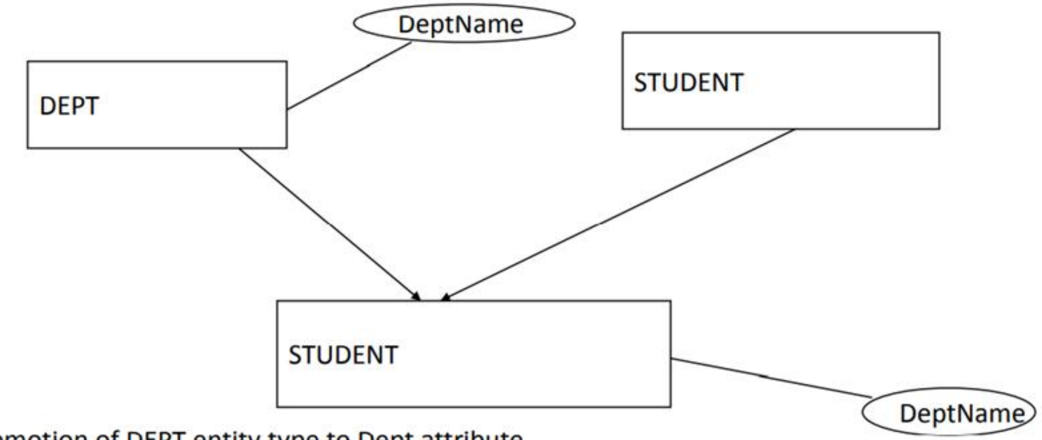
- An attribute that exists in several entity types may be elevated or promoted to an independent entity type.
- **Eg:**




Promotion of Dept Attribute to DEPT entity type

Instead of repeating the Department attribute in each entity, it becomes its own entity (DEPARTMENT).Relationships are then created between DEPARTMENT and the other entities (STUDENT, INSTRUCTOR, and COURSE).Additional attributes or details for DEPARTMENT can be added later as needed.

- An inverse refinement to the previous case may be applied.
- **Eg:** If the DEPARTMENT entity in the initial design has only one attribute (Dept_name) and is related to only the STUDENT entity, it can be simplified. Instead of keeping DEPARTMENT as a separate entity, it can become an attribute of STUDENT (**Eg:** each student has a department).



Demotion of DEPT entity type to Dept attribute